AIAC Assignment 19.1

RAVITEJA VADLURI
2303A51942
BATCH – 12

Task 1 – Python to Java Conversion Task:

Translate a simple Python class into Java using AI assistance.

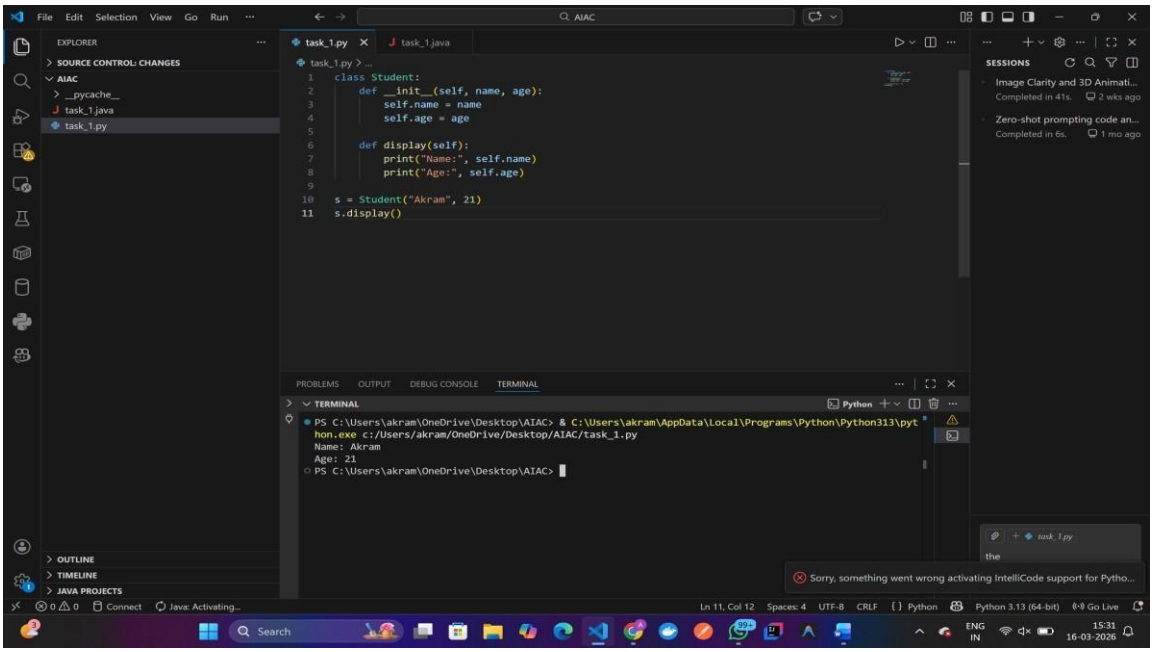
Instructions:

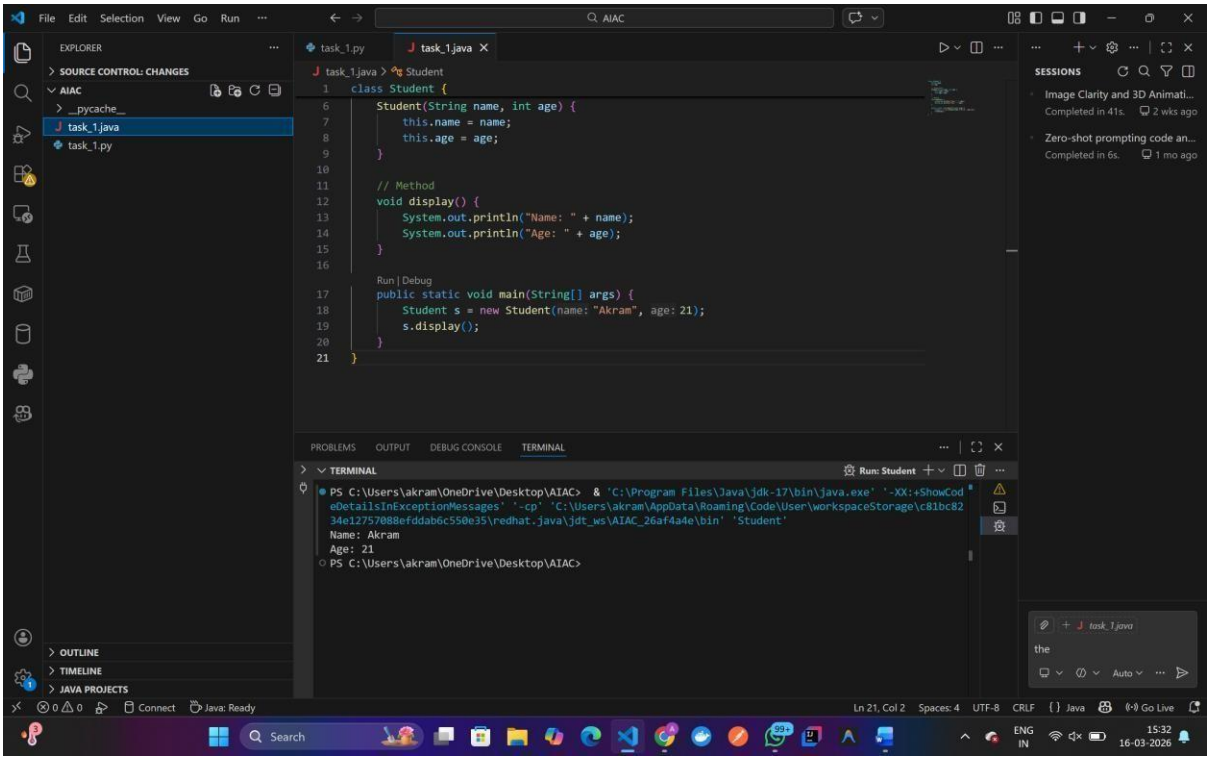
- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

CODE:





Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

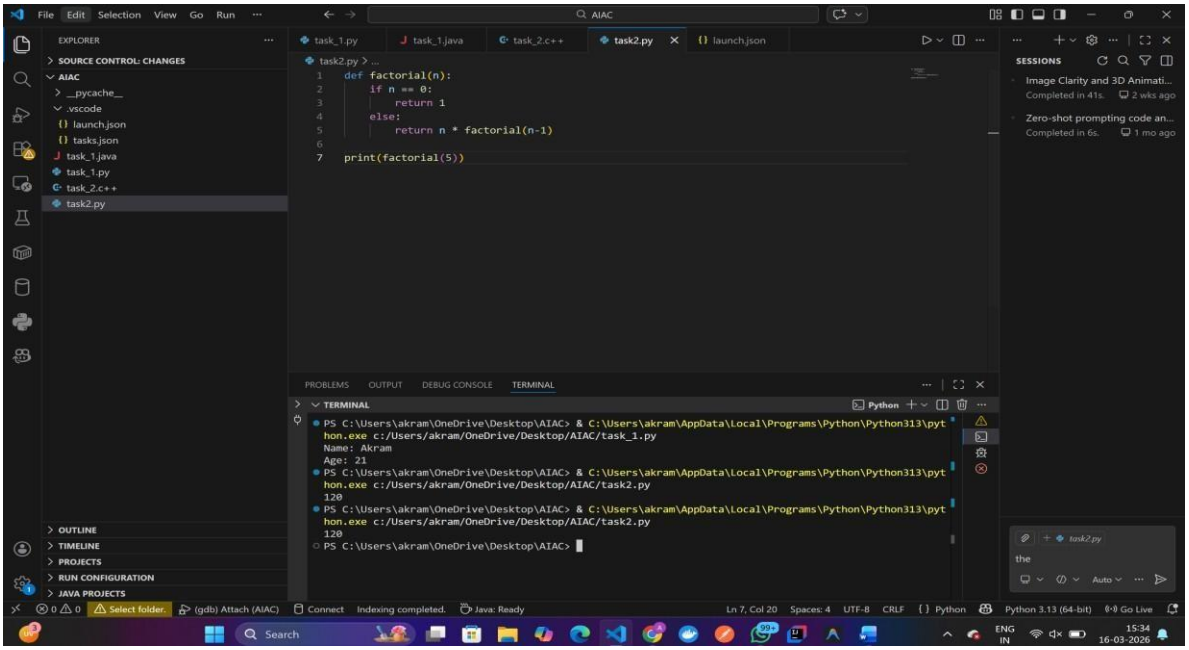
Instructions:

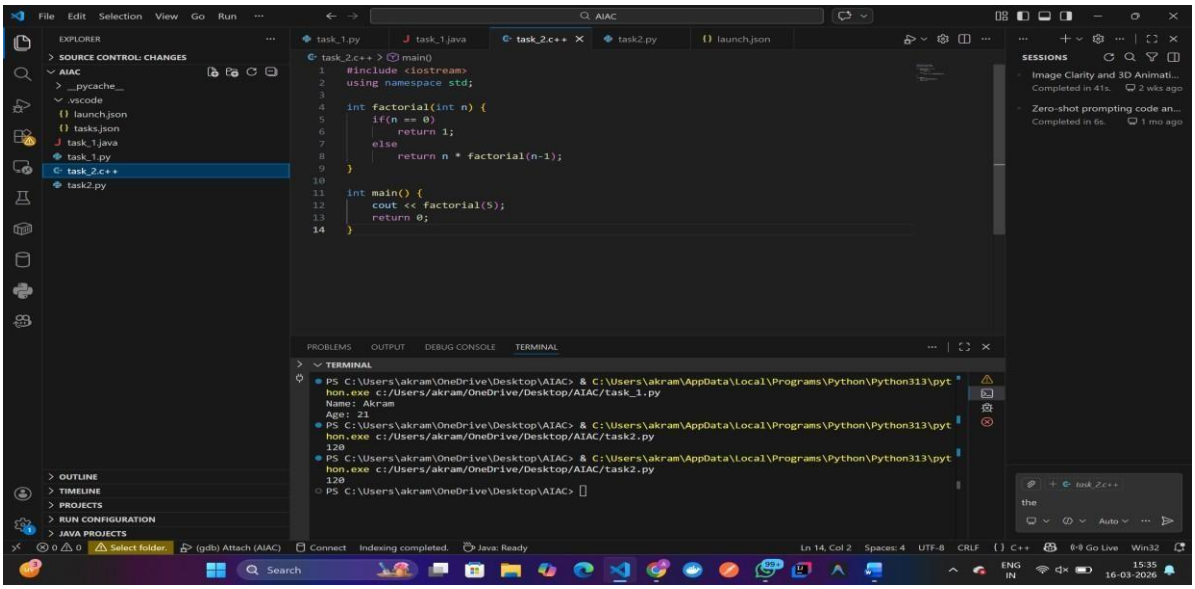
- Ask AI to translate a recursive C++ factorial function into Python.
- Validate correctness by testing with sample inputs.
- Ensure Python version follows Pythonic conventions.

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

CODE:





Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

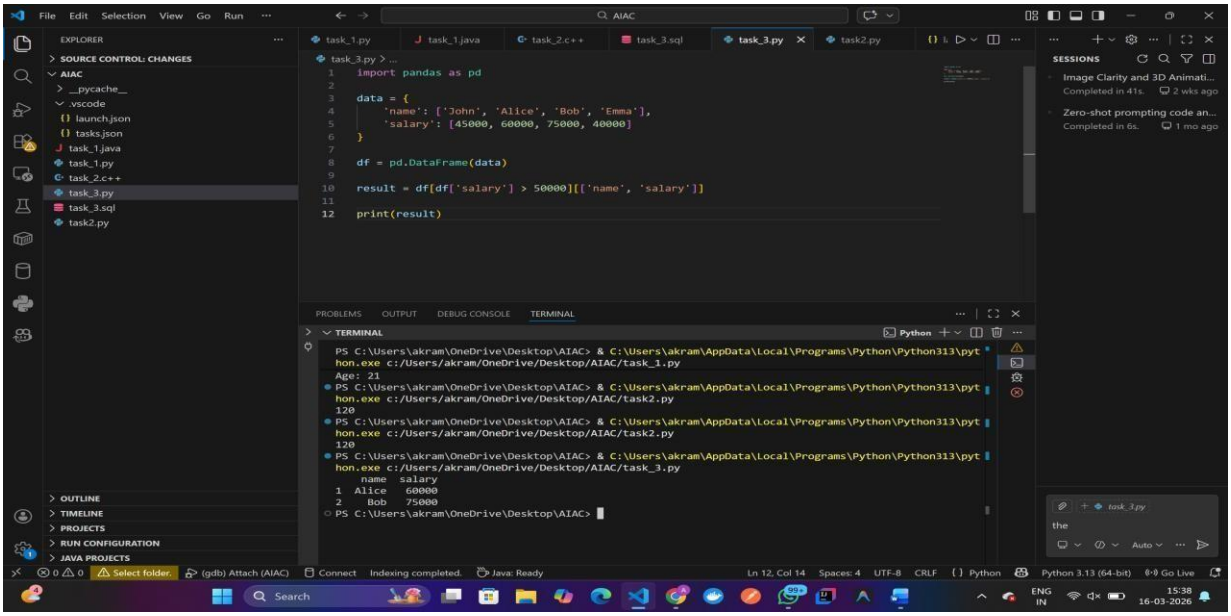
Instructions:

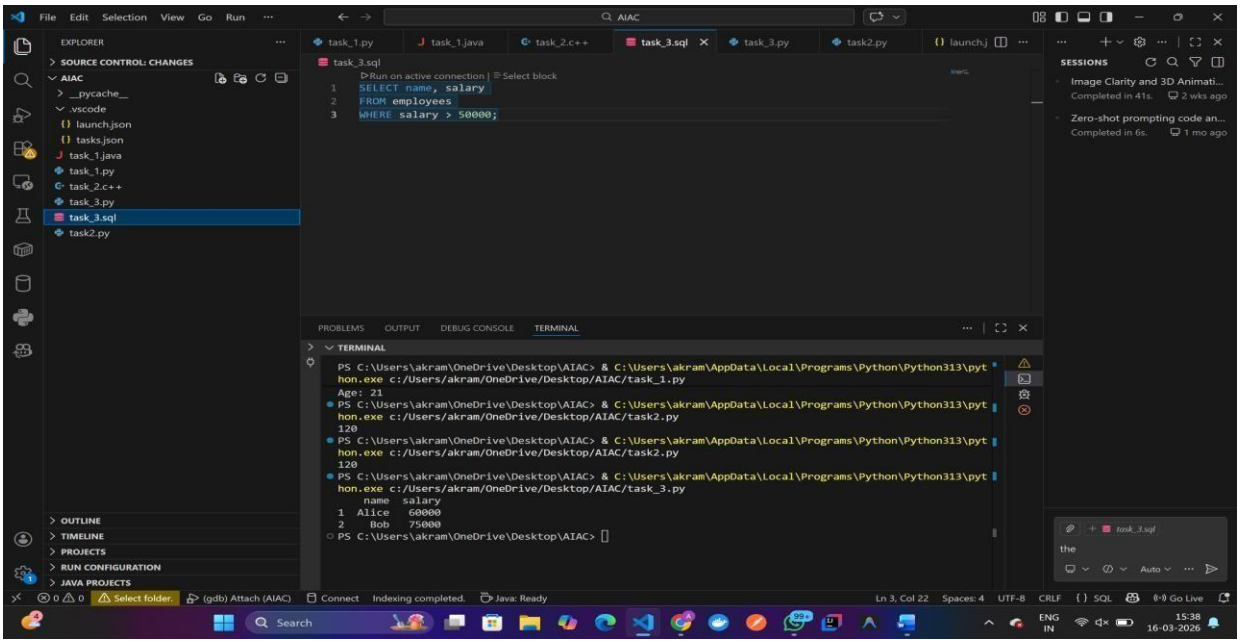
- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

CODE:





Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

- Converting a Python web scraping snippet into JavaScript for browser automation.
- Translating a Java method into C# for enterprise applications.

Instructions:

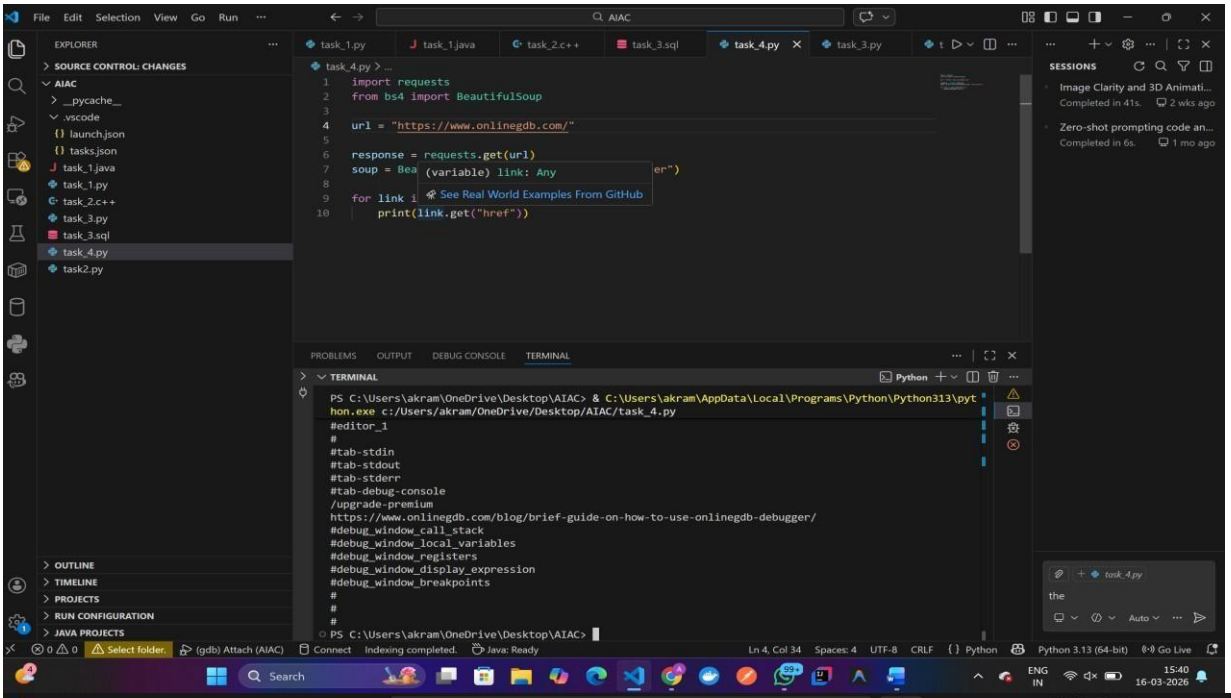
- Prompt AI to translate code between two chosen languages.
  - Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax,

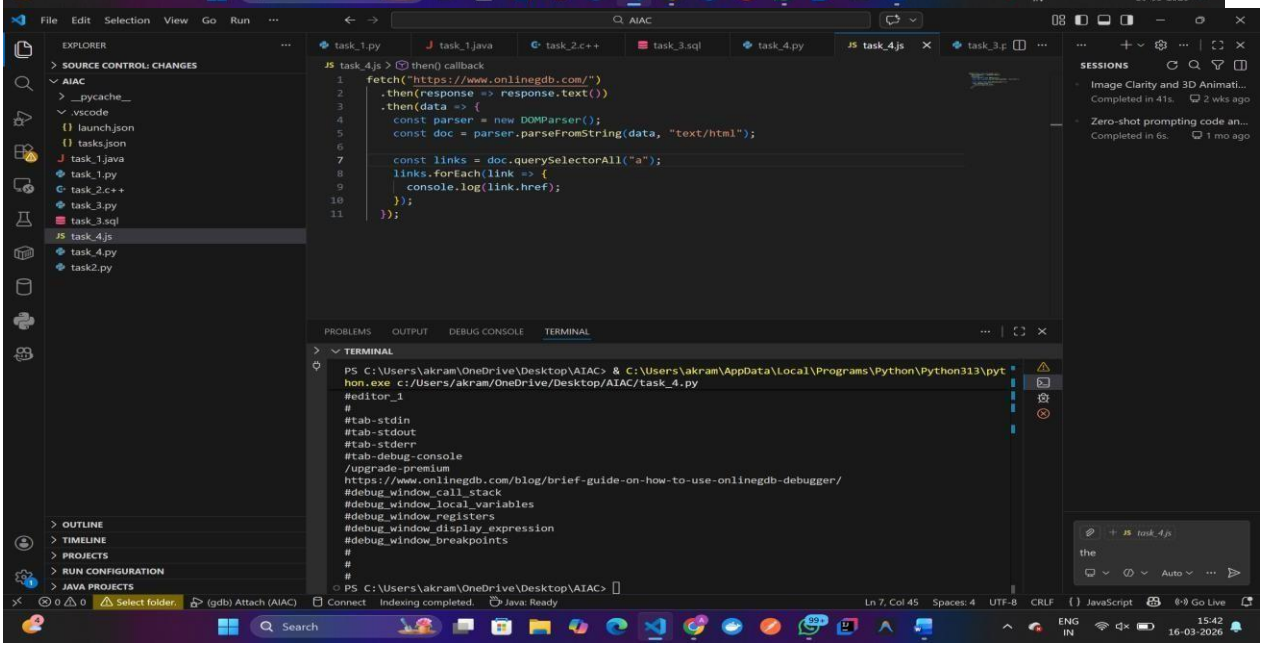
libraries).

Expected Output:

- Working equivalent code snippets in at least two different

programming languages CODE:





End Lab Observations

AI-assisted tools significantly simplify the process of code translation between programming languages such as Python, Java, C++, and SQL to Pandas.

While translating code, syntax differences (e.g., indentation in Python vs. braces in Java/C++) and data type handling were observed as key challenges.

AI-generated code is generally correct, but manual verification and testing are essential to ensure functional accuracy.

In Python-to-Java conversion, constructors, class structures, and type declarations required careful validation.

During C++ to Python conversion, it was observed that Python offers simpler and more concise syntax, especially for recursive functions like factorial.

Translating SQL queries to Pandas operations highlighted the difference between declarative (SQL) and procedural (Python/Pandas) approaches.

Real-time multi-language conversion demonstrated that libraries and runtime environments differ, which may require additional adjustments beyond direct translation.

Debugging AI-generated code helped in understanding semantic differences between programming languages.